Lighthouses: Guardians of the Coast

A lighthouse is a tall tower that carries a powerful light to help sailors find their way. Its main jobs are to mark dangerous places such as rocks and shoals, and to show ships where the coast lies. Because every lighthouse has its own pattern, a sailor can identify which one is in sight.

The most famous ancient lighthouse was the Pharos of Alexandria, built on an island off the coast of Egypt in the third century BCE. It was counted among the Seven Wonders of the Ancient World. Its light came from a fire, and the tower stood for many centuries before earthquakes eventually brought it down.

For most of history, lighthouses burned wood, coal or candles, and later oil lamps. These flames were dim, so builders placed them as high as they could. The invention of better lamps and of reflectors made lights much brighter, but the real breakthrough came with the lens.

In 1822 the French engineer Augustin Fresnel designed a new kind of lens made of many glass rings and prisms. Earlier lenses were thick, heavy and absorbed much of the light. The Fresnel lens bends light from the lamp into a narrow, powerful beam while using far less glass, and it allowed a light to be seen from much greater distances.

Many lighthouses appear to flash because the lens assembly rotates around the lamp. As the beam sweeps past an observer, they see a brief flash followed by darkness. By changing the number of lenses or the speed of rotation, builders gave each lighthouse a distinct pattern, called its characteristic, such as one flash every ten seconds.

Lighthouse towers are shaped to survive wind and waves. Many are round and tapered, wide at the base and narrower at the top, because a curved surface lets wind and water slide around it. Walls are very thick and built from stone, brick or concrete. Some stand on rocks that are covered by the sea at high tide.

Building a lighthouse on an exposed rock out at sea was a great engineering challenge. Crews could often work only in calm weather during the summer, and the sea might wash away a half-finished structure. The Eddystone lighthouse off the coast of England was rebuilt several times, and one famous version designed by John Smeaton in the eighteenth century used interlocking blocks of granite.

Lighthouses were once staffed by keepers who lived at or near the tower. A keeper trimmed wicks, cleaned the lens, wound the clockwork that turned it and kept a log of weather and passing ships. Keepers also climbed the stairs many times a day, since the lamp had to be tended through the night.

During the twentieth century, lighthouses were gradually automated. Electric lamps replaced oil, and sensors and timers switched the light on at dusk. Automation meant that most lighthouses no longer needed a resident keeper, and many keepers' cottages were closed or converted to other uses.

Light is not the only warning a lighthouse can give. In fog, a light may be hard to see, so many stations also sounded a signal such as a horn, a bell or a siren. Sailors learned to listen for these sounds when visibility was poor.

Modern navigation relies on satellite positioning systems, radar and electronic charts, so ships need lighthouses less than before. Even so, lighthouses remain useful as a backup when electronics fail and as reliable landmarks for small boats. Their lights are also given a standard colour and range on nautical charts.

Many old lighthouses have found new lives. Some are museums that tell the story of the coast, some are lodgings for visitors, and others are preserved as historic landmarks. Communities often work to protect them, because the towers have become symbols of safety and of a seafaring past.

Lightships served as floating lighthouses where building a tower was not possible. A lightship was a sturdy vessel anchored over a hazard such as a sandbank, with a lamp raised on a mast. Crews lived aboard for weeks at a time, and the ships had to withstand storms while staying firmly in place.

Lighthouse designers also had to think about how the light would be seen. Lamps were placed at a height that matched the distance at which the light needed to be visible, since the curve of the Earth hides low lights from far-off ships. A taller tower therefore gave a longer range, which is why the tallest lighthouses stand on headlands and cliffs.